

A multi-modal audit of Alzheimer's disease blood transcriptomics

A cell-composition account of the blood-RNA reproducibility paradox

Paradox

Study A

Study B

Study C



different genes

similar AD
classification

Why?

Audit

~1,350 participants

3 bulk cohorts

ADNI validation

scRNA-seq

TCR repertoire

Interpretable ML

5 modalities

→ same answer

Cell-composition account

Blood RNA



Immune-cell
composition

lymphocyte ↓



monocyte ↑



neutrophil ↑

Different panels

same phenotype

Implications

Requirements for
credible discovery

- Symmetric composition
adjustment
- Donor-level
single-cell statistics
- Composition-aware
benchmarking

Key findings

67%

biomarkers lost
after de-confounding

88% → 48%

replication collapses
(symmetric adjustment)

873 → 6

scRNA DE genes
(donor-level stats)

0.63 ≈ 0.63

composition vs panel
(AUC, head-to-head)